Hair mercury levels of women of reproductive age in Ontario, Canada: implications to fetal safety and fish consumption.

OBJECTIVE: To study hair mercury concentrations among women of reproductive age in relation to fish intake in Ontario, Canada. STUDY DESIGN: Three groups were studied: 22 women who had called the Motherisk Program for information on the reproductive safety of consuming fish during pregnancy, a group of Japanese residing in Toronto (n=23) consuming much larger amounts of fish, and a group of Canadian women of reproductive age (n=20) not seeking advice, were studied. Mercury concentrations in hair samples were measured using inductively coupled plasma mass spectrometry. Seafood consumption habits were recorded for each participant. Based on the types of fish consumed and consumption frequencies, the estimated monthly intake of mercury was calculated. Hair mercury concentrations were correlated to both the number of monthly seafood servings and the estimated ingested mercury dose. RESULTS: There were significant correlations between fish servings and hair mercury (Spearman $r=0.73$, $P<.0001$) and between amounts of consumed mercury and hair mercury concentrations (Spearman $r=0.81$, $P<.0001$). Nearly two thirds of the Motherisk callers, all of the Japanese women, and 15% of the Canadian women of reproductive age had hair mercury above 0.3 microg/g, which was shown recently to be the lowest observable adverse effect level in a large systematic review of all perinatal studies. CONCLUSIONS: Because of very wide variability, general recommendations for a safe number of fish servings may not be sufficient to protect the fetus. Analysis of hair mercury may be warranted before pregnancy in selected groups of women consuming more than 12 ounces of fish per week, as dietary modification can decrease body burden and ensure fetal safety.